Hackathon Challenge Pack

Common challenge mechanics

- Level 1 (0 - 3 hours): demonstrate a working end-to-end core journey.
- Judging: award points for observable behaviour, not architecture slides or unsupported AI claims. Use synthetic or properly anonymised data only. No production credentials, real customer PII, external payments or live collection calls.
- Organiser responsibility: provide schemas, sample data, hidden tests, credentials for sandbox APIs, evaluation rubric and expected policy/regulatory rules. A prototype is not certified for regulatory compliance.
- No AI resources provided: the organiser will not provide AI model access, API keys, usage credits, model hosting or speech services. Teams must use their own accounts and bear any associated costs.

1. Digital Fraud Investigation System

Category: AI for Fraud Detection, Investigation & Financial Crime Prevention
Applicable Domains: Digital Lending, Insurance, Core Banking and Digital Financial Services
Focus: Fraud Detection, Risk Intelligence, Investigation, Prevention and Case Management

Problem Statement

Financial institutions increasingly rely on digital applications to onboard customers, process loan applications, issue insurance policies, execute financial transactions and deliver banking services. While these digital journeys improve customer experience and operational efficiency, they also introduce opportunities for fraud.

Fraud is no longer limited to suspicious financial transactions. It can occur at different stages of a customer's digital journey, from identity verification and account creation to loan disbursement, insurance claims and fund transfers. Fraudsters may use stolen identities, manipulated documents, compromised accounts, multiple fake profiles, shared devices, coordinated transactions or fabricated insurance claims to exploit financial systems. For financial institutions operating across lending, insurance and core banking, detecting these activities independently within each system is often insufficient. A transaction may appear legitimate in isolation but become suspicious when connected to a customer's previous applications, login behaviour, device history or activities across other financial products.

For example, a customer may apply for multiple loans using different identities but the same device, receive disbursements into related bank accounts and subsequently initiate unusual transfers. Each activity may appear acceptable individually, but the combined behaviour could indicate coordinated fraud.

The challenge is to build an AI-powered Digital Fraud Investigation System that identifies suspicious activities across digital financial journeys, connects related events, assesses fraud risk and enables financial institutions to investigate and respond appropriately. The application must go beyond basic fraud detection based on transaction amounts or predefined thresholds. It should identify suspicious patterns across customer profiles, transactions, digital applications, devices, login activities, lending operations, insurance activities and core banking systems.

The solution must distinguish potentially fraudulent behaviour from legitimate customer activity, explain why an event was flagged, retain supporting evidence and enable authorised users to investigate, escalate or dismiss cases.

Participants are encouraged to introduce additional fraud detection techniques and innovative approaches beyond the specified scenarios. Solutions demonstrating meaningful detection of emerging or sophisticated digital fraud patterns will receive additional consideration within the evaluation framework.

The organiser will assess actual application behaviour rather than a static dashboard, predefined presentation or standalone AI-generated response.

Digital Fraud Scenarios to Be Addressed

The solution should consider the following types of digital fraud. These scenarios represent practical challenges faced by financial institutions and provide the basis for progressive evaluation.

1. Identity Fraud and Synthetic Customer Profiles

Fraudsters may create customer profiles using stolen identification details, altered documents or combinations of genuine and fabricated information. The application should identify inconsistent identity attributes, repeated contact information, suspiciously similar profiles and unusual relationships between supposedly independent customers.

2. Device Sharing and Coordinated Fraud

Multiple customers may use the same device, IP address, contact number or digital identity attributes to submit applications or initiate transactions. The system should identify potentially coordinated activity while recognising legitimate shared devices, such as family members using a common phone.

3. Account Takeover and Suspicious Login Activity

Fraudsters may gain unauthorised access to customer accounts through stolen credentials, phishing or compromised authentication mechanisms. Suspicious behaviour may include repeated failed login attempts, unexpected device changes, unusual access locations, password resets followed by large transfers or changes to beneficiary information.

4. Cross-System and Organised Fraud

Fraud patterns may become visible only when information from lending, insurance and core banking systems is connected. The solution should correlate activity across these systems and identify related customers, accounts, devices, applications, transactions and claims that could indicate organised fraud.

Level 1: Detect, Analyse and Flag Suspicious Digital Activity
Suggested time: 0 -3 hours
Objective: Build a working fraud detection application that ingests digital financial activity, identifies suspicious patterns and presents explainable fraud alerts for investigation.

Required Capabilities
1. Data Ingestion and Integration
Build an application that allows users to upload or select data from different digital financial systems. The application should support the following sample data:

- Customer and account profiles containing customer identifiers, account details, contact information and onboarding information.
- Transaction batches containing amounts, timestamps, sender and beneficiary details, transaction channels and statuses.
- Device and login logs containing device identifiers, IP addresses, login timestamps, authentication outcomes and device changes.

The application should accept common structured formats such as CSV, Excel and JSON. It must validate uploaded records, identify missing mandatory information and avoid silently ignoring invalid data. Participants should use a common data structure that allows records from different systems to be linked through customer, account, application, device or transaction identifiers.
2. Fraud Detection Rules and Behavioural Analysis
Implement a detection mechanism that identifies suspicious activities using configurable rules, statistical analysis, machine learning or a combination of approaches. At a minimum, the application must detect the following four scenarios:

Scenario	Expected detection behaviour
Suspicious transfers	Identify unusual amounts, rapid transfers, new beneficiaries or transactions inconsistent with customer history
Shared devices	Identify multiple customer profiles using the same device and assess whether the relationship is suspicious
Repeated login failures	Detect repeated unsuccessful logins, particularly when followed by a successful login or sensitive account activity
Legitimate lookalikes	Avoid incorrectly classifying ordinary customer behaviour as fraud merely because it resembles a suspicious pattern

Participants should also implement at least one additional fraud scenario from the challenge catalogue, such as suspicious loan applications or duplicate insurance claims.

3. Fraud Risk Assessment
Every detected suspicious activity should receive a risk classification.
The application may use a risk score, such as 0–100, or clearly defined categories such as Low, Medium and High. The score should consider multiple relevant indicators rather than relying only on a single threshold.
For example, a high-value transfer should not automatically be considered fraudulent if the customer regularly performs similar transactions.
However, a high-value transfer immediately following multiple failed login attempts and a new-device login may warrant greater scrutiny. The application must explain the factors contributing to the assigned risk level.
4. Fraud Alert Dashboard
Develop a dashboard where users can inspect detected suspicious activities.
The dashboard should display the number of processed records, flagged events, risk distribution and fraud categories. Users must be able to filter alerts by customer, transaction, product, risk level, date and detection type.

Selecting an alert should display the underlying event, the reasons it was flagged and the supporting records.

5. Basic Fraud Case Queue

Each actionable fraud alert should be available in a case queue.

A case should contain a unique identifier, affected customer or account, suspected fraud category, risk level, supporting evidence and investigation status. Users should be able to open the case and record an initial decision, such as Requires Investigation or False Positive. The system must preserve the original alert even when its classification is subsequently changed.

Expected Submission

A working application demonstrating data ingestion, fraud detection, risk assessment, an alert dashboard and a basic investigation queue.

2. Regulatory Change Impact Agent

Category: AI for Regulatory Intelligence, Compliance & BFSI
Applicable Regions: Saudi Arabia (KSA), United Arab Emirates (UAE), India and selected African countries
Regulatory Focus: Artificial Intelligence, Banking, Financial Services and Insurance (BFSI)

Problem Statement

Financial institutions operate in a constantly evolving regulatory environment. Governments, central banks, financial regulators and other supervisory authorities regularly publish new regulations, circulars, amendments, guidelines and compliance requirements.

These changes may introduce new obligations, modify existing regulations, extend compliance deadlines, withdraw earlier instructions or introduce requirements that affect how financial institutions operate.

Today, compliance teams often monitor regulatory websites manually, download circulars, compare them with earlier regulations, interpret the changes and identify which internal policies, processes, systems and controls need to be updated. This process is time-consuming, particularly for organisations operating across multiple countries.

The challenge is to build an AI-powered Regulatory Change Impact Agent that continuously monitors regulatory publications across KSA, UAE, India and selected African countries, understands how new regulations relate to existing ones, and identifies the changes organisations need to make to remain compliant.

The solution should go beyond summarising regulatory announcements. It must establish relationships between current and historical regulations, identify new or modified obligations, assess their impact on internal policies and controls, and recommend the necessary changes.

For example, if a regulator introduces a new requirement governing the use of AI in credit decisioning, the system should identify the previous regulations governing that activity, determine what has changed, identify affected internal credit policies and AI governance controls, and recommend specific modifications. The application must also support human review, approvals, accountability and traceability so that compliance teams can confidently act on its recommendations.

Level 1: Monitor, Discover and Understand Regulatory Changes
Suggested time: 0 - 3 hours
Objective: Build a working regulatory intelligence application capable of discovering regulatory publications, extracting their contents and identifying changes against previously published regulations.

Required Capabilities
1. Regulatory Source Monitoring
Build a monitoring mechanism that can retrieve regulatory publications from official government and regulatory authority websites across the defined regions.
The application should support configurable regulatory sources. Examples include SAMA and other relevant Saudi authorities, the Central Bank of the UAE, the Reserve Bank of India, and selected African central banks or financial regulators. Participants must demonstrate monitoring of at least four configured sources, representing the four geographic regions. The system should:

- Retrieve newly published regulations, circulars, amendments, consultation papers and guidelines.
- Identify the issuing authority, jurisdiction, publication date, document title, reference number and effective date, wherever available.
- Distinguish newly published documents from documents already processed.
- Allow users to configure monitoring frequency and manually trigger a monitoring cycle.
- Maintain a searchable repository of all discovered regulatory publications.
- Support manual upload of PDF, DOCX or text-based circulars when automatic retrieval is unavailable.

2. Regulatory Document Understanding
Once a document is retrieved or uploaded, the system should analyse its contents and extract relevant regulatory information. The application should identify:

- Regulatory obligations and mandatory requirements.
- Applicability to AI, banking, lending, insurance, payments or other BFSI activities.
- Compliance deadlines and effective dates.
- References to earlier regulations or circulars.
- Clauses that introduce, amend, replace or withdraw existing requirements.
- Entities, departments or regulated activities to which the requirements apply.

The extracted information should be presented in a structured format rather than only as an AI-generated summary. Each extracted obligation must include a reference to the relevant clause or section of the source document.

3. Historical Regulation Comparison
The application must compare newly identified regulations with previously stored regulatory publications. For example, when a new circular amends an existing requirement, the system should retrieve the earlier circular and identify the relevant differences. It should classify changes such as:

- New obligation introduced.
- Existing obligation modified.
- Obligation withdrawn or superseded.
- Compliance deadline extended or shortened.
- Additional clarification issued.
- No material change identified.

The system should display the previous and updated regulatory clauses side by side, highlighting the changes and explaining their significance in simple English.
4. Regulatory Intelligence Dashboard
Develop a dashboard where users can view regulatory developments by country, regulator, publication date, business area and change classification.

The dashboard should distinguish new publications, amendments, items awaiting review and documents for which the system could not establish a reliable relationship with earlier regulations. Users should be able to open any publication, inspect its extracted obligations and review its source references.

Expected Submission
A working web application demonstrating source monitoring, document ingestion, obligation extraction, historical comparison and a regulatory dashboard.

3. Offer & Product Recommendation Engine

Category: AI for Customer Intelligence, Personalised Financial Offers & Revenue Growth
Applicable Domains: Digital Lending, Core Banking, Insurance and Digital Financial Services
Focus: Income Estimation, Customer Profiling, Product Eligibility, Cross-Selling, Upselling and Conversational AI Assistance

Problem Statement

Financial institutions offer a wide range of financial products, including personal finance, vehicle finance, home finance, credit cards, deposits, insurance and other financial services. However, customers generally approach a bank or financial institution with a specific requirement, without necessarily knowing what other products, benefits or financial offers they may be eligible for.

For example, a customer visiting a digital lending application to apply for personal finance may also qualify for a higher financing amount, a credit card, vehicle finance or an insurance product. The financial institution may already have sufficient information about the customer's income, financial behaviour, repayment history and creditworthiness to identify these opportunities. However, this information is often distributed across banking transactions, credit bureau records, lending systems and product catalogues.

Traditional product recommendation approaches frequently rely on broad customer segmentation, static marketing campaigns or simple eligibility filters. These approaches may overlook a customer's actual financial capacity, changing needs and eligibility for products beyond the one originally requested. The challenge is to build an AI-powered Offer & Product Recommendation Engine that understands a customer's financial profile, estimates income from transaction behaviour, evaluates creditworthiness and product eligibility, and recommends the most relevant financial products and offers. The solution should support both existing customers, whose transaction and financial history may already be available, and new customers, who may have limited financial information.

The recommendation engine must go beyond suggesting the product a customer originally requested. It should identify suitable opportunities for cross-selling and upselling while ensuring that every recommended offer meets the financial institution's eligibility criteria and affordability requirements.

For example, a customer applying for personal finance of ₹5 lakh may be eligible for ₹8 lakh based on verified income, existing obligations and the institution's lending policies. The application should explain this opportunity while also identifying other suitable products, such as a credit card or vehicle finance, based on the customer's financial profile.

Similarly, a customer with significant monthly savings and regular account balances may be a suitable candidate for a deposit or investment-related product, provided the relevant product eligibility and suitability rules are met. The solution must not recommend products merely because they are commercially attractive to the financial institution. Recommendations should be relevant, explainable, affordable and supported by available customer information.

The application should also include a small AI-powered conversational assistant that allows customers or relationship managers to ask questions about eligibility, available offers, financing capacity and alternative products. The assistant must operate using the same underlying eligibility and recommendation engine as the application, ensuring that conversational answers are consistent with the actual offers displayed.

Participants are encouraged to introduce innovative recommendation approaches that improve customer engagement, financial product discovery, cross-selling, upselling and conversion opportunities.

The organiser will assess actual application behaviour rather than a static dashboard, predefined presentation or standalone AI-generated response.

Product Recommendation Scenarios to Be Addressed

The solution should consider the following customer scenarios. These represent practical opportunities for financial institutions to improve product discovery and customer engagement.

1. Personal Finance Eligibility and Upselling

A customer applies for a personal finance product of x amount. Based on estimated income, existing financial obligations, credit bureau information and product eligibility rules, the system determines that the customer may qualify for a higher financing amount.

The application should identify the eligible amount, explain the basis of the calculation and present the additional financing opportunity without encouraging borrowing beyond the customer's assessed repayment capacity.

2. Cross-Selling Across Financial Products

A customer applying for personal finance may also qualify for a credit card, vehicle finance, insurance or another financial product.

The system should evaluate the customer's profile against the available product catalogue and identify additional eligible offers.

Recommendations must consider financial capacity, product suitability and the institution's business rules rather than relying solely on the product initially requested.

3. Income Estimation from Bank Transactions

A customer provides bank statements or transaction histories containing salary credits, business receipts, internal transfers, refunds and irregular income. The application must identify genuine income sources, distinguish them from non-income credits and estimate a sustainable monthly income.

For example, a transfer between the customer's own accounts must not be counted as additional income merely because it appears as a credit transaction.

4. Customers with Irregular or Variable Income

Self-employed customers, freelancers and business owners may not receive a fixed salary every month. The system should evaluate recurring receipts, income consistency, seasonal variations and available historical information to estimate income appropriately. It should avoid rejecting a customer solely because the transaction history does not contain a conventional monthly salary credit.

5. Thin-File and New-to-Bank Customers

Some customers may have limited transaction history or insufficient credit bureau information. The system should recognise missing information, identify what can reasonably be assessed and request additional evidence where necessary. It must not invent credit scores, income values or eligibility decisions.

6. Product Ineligibility and Alternative Recommendations

A customer may request a product for which they do not meet the eligibility criteria. For example, a requested loan amount may exceed the maximum permitted debt burden, or a premium credit card may require a higher verified income. The application should explain the exclusion clearly and, where possible, recommend a suitable alternative.

7. Conversational Product Discovery

A customer or relationship manager may ask:

- "Am I eligible for a higher personal finance amount?"
- "What other products can I apply for?"
- "Why am I not eligible for this credit card?"
- "What happens if my monthly income increases?"

The assistant should answer using the customer's assessed financial information and the current product catalogue. It should also allow users to navigate to relevant offers and understand the calculations supporting the recommendations.

Level 1: Understand Customer Financial Profile and Determine Product Eligibility
Suggested time: 0 - 3 hours
Objective: Build a working application that ingests customer financial information, estimates income, evaluates product eligibility and presents eligible financial products with clear explanations.

Required Capabilities
1. Customer Financial Data Ingestion
Build an application that allows users to upload or select customer financial information from synthetic datasets. The application should support:

- Bank transaction histories containing transaction dates, descriptions, amounts, debit/credit indicators, account identifiers and balances.
- Customer profiles containing age, employment category, occupation, declared income and existing banking relationships.
- Credit bureau attributes containing credit scores, existing loans, repayment history, outstanding balances and monthly obligations.
- A financial product catalogue containing product types, financing limits, interest or profit rates, tenures, eligibility conditions and available offers.
- Product eligibility rules containing income thresholds, debt burden limits, credit score requirements, age restrictions and other relevant criteria.

The application should accept CSV, Excel and JSON formats. It must validate uploaded information, identify missing or inconsistent values and prevent invalid records from silently influencing financial calculations. The system should maintain a clear relationship between the customer profile, financial transactions, bureau attributes and available products.
2. Income Identification and Estimation
Develop an income estimation mechanism that analyses transaction history and identifies genuine income.
The system should distinguish salary credits, recurring business receipts, other identifiable income sources, internal account transfers, refunds, loan disbursements and unrelated credits. For example, if a customer receives a monthly salary of ₹80,000 and transfers ₹30,000 between their own accounts, the system should not incorrectly calculate monthly income as ₹1,10,000.
The application should estimate a sustainable monthly income using available transaction history. For customers with irregular income, it should consider multiple months of receipts, consistency and unusual fluctuations rather than assuming that the highest monthly credit represents regular income. Every estimated income value must include a calculation breakdown and supporting transactions.
3. Customer Financial Capacity Assessment
Use estimated income and credit bureau attributes to calculate the customer's indicative financial capacity. The assessment should consider existing monthly loan repayments, other relevant financial obligations, credit history and applicable debt burden or affordability rules. The application should calculate an indicative amount available for additional monthly repayment obligations and use this when evaluating financing products. Where relevant information is unavailable, the system should mark the assessment as incomplete rather than inventing values.
4. Product Eligibility Evaluation
Build a rule-based eligibility engine that evaluates the customer against the supplied product catalogue. The engine should determine which products the customer qualifies for based on the institution's eligibility requirements. At a minimum, participants must demonstrate eligibility assessment for:

Product	Expected eligibility assessment
Personal finance	Income, existing obligations, creditworthiness, financing amount and tenure

Vehicle finance	Income, repayment capacity, product limits and financing conditions
Credit card	Income threshold, credit history, existing obligations and card category
Deposit or savings product	Account eligibility, minimum balance or deposit requirements
Insurance product	Relevant customer attributes, product eligibility and applicable coverage conditions

The application should distinguish between eligible, conditionally eligible, ineligible and insufficient-information outcomes.

5. Product Eligibility Dashboard

Develop a customer-facing or relationship-manager-facing dashboard that displays the assessed customer profile. The dashboard should show estimated monthly income, existing financial obligations, indicative financing capacity and available financial products. Each product should display its eligibility status, applicable offer details and the primary reasons supporting the decision. Ineligible products should not simply disappear. Users should be able to understand why a particular product was excluded.

6. Basic Recommendation Explanation

For every eligible product, provide a simple explanation of why it is suitable. For example: "Based on your estimated monthly income, existing financial obligations and the current eligibility rules, you may qualify for personal finance up to ₹8 lakh." The application must display the assumptions and eligibility rules used to reach the result. Recommendations must be calculated using the actual uploaded data and product rules, not generated as unsupported AI statements.

Expected Submission

A working application demonstrating customer financial data ingestion, income estimation, affordability assessment, product eligibility evaluation and a customer recommendation dashboard.

4. Early Warning System (EWS)

Category: AI for Credit Risk Intelligence, Portfolio Monitoring & Proactive Risk Management

Applicable Domains: Digital Lending, Loan Management, Core Banking, SME Finance and Financial Services

Focus: Early Risk Identification, Repayment Behaviour, Delinquency Prediction, Portfolio Intelligence, Customer Monitoring and Preventive Interventions

Problem Statement

Financial institutions continuously manage the risk of customers failing to meet their financial obligations. While traditional credit risk management focuses on assessing customers before approving a loan, the customer's financial circumstances and repayment behaviour may change significantly after disbursement.

A customer who was considered financially stable at the time of loan approval may subsequently experience declining income, irregular cash flow, increasing debt obligations, missed repayments or financial distress. These changes may occur gradually and may not immediately result in a loan becoming delinquent. Traditional monitoring systems often rely on predefined indicators such as Days Past Due (DPD), missed instalments or overdue balances. While these indicators are important, they may identify risk only after repayment difficulties have already started.

Similarly, a small business may experience declining sales receipts, increasing payment obligations and irregular cash flows before it misses a loan instalment. Financial institutions need to identify such warning signs early, understand their significance and initiate appropriate preventive measures before the situation deteriorates.

The challenge is to build an AI-powered Early Warning System that continuously evaluates customer repayment behaviour, financial exposure and emerging risk signals to identify potential credit deterioration and recommend timely interventions. The application should go beyond simply flagging overdue loans. It must analyse multiple indicators, recognise changes in customer behaviour, distinguish temporary irregularities from sustained financial deterioration and provide an explainable assessment of emerging credit risk.

The system should support both individual customer monitoring and portfolio-level risk analysis. It should help credit risk teams, relationship managers and collections teams understand which customers require attention, why their risk is increasing and what actions may be appropriate.

Early Warning Systems cover a broad range of risks. Participants are encouraged to introduce additional risk indicators, predictive techniques and monitoring capabilities beyond the mandatory scenarios. Solutions demonstrating meaningful innovation in identifying emerging financial stress or portfolio deterioration will receive additional consideration within the evaluation framework.

The organiser will assess actual application behaviour rather than a static dashboard, predefined presentation or standalone AI-generated response.

Early Warning Scenarios to Be Addressed

The solution should consider the following risk scenarios. These represent practical situations in which financial institutions may benefit from identifying financial deterioration before significant delinquency or default occurs.

1. Repayment Behaviour Deterioration

A customer who previously made regular repayments begins paying instalments late, making partial payments or missing scheduled obligations. The system should identify changes in repayment consistency, increasing Days Past Due, repeated missed instalments and other repayment irregularities. It should distinguish an isolated payment delay from a recurring pattern of deterioration.

2. Declining Income and Cash Flow

A customer or business experiences a gradual decline in income or account inflows. For example, an SME customer whose average monthly business receipts decline over three consecutive months may be experiencing financial pressure even though loan repayments remain current. The system should identify sustained deterioration in cash flow, income consistency and available financial capacity.

3. Increasing Financial Obligations and Exposure

A customer takes additional credit facilities, increases utilisation of existing limits or accumulates higher repayment obligations. The application should assess whether the customer's overall financial exposure is increasing faster than their ability to repay. It should consider outstanding balances, existing obligations, available limits and exposure trends.

4. Temporary Missed Payments and Legitimate Exceptions

A customer misses one repayment due to a bank holiday, processing delay, temporary salary delay or another isolated event. The system should recognise that not every missed payment indicates significant financial deterioration. It should consider historical repayment behaviour, subsequent payment recovery and the availability of supporting evidence before assigning a warning level.

5. Early Delinquency and Roll-Forward Risk

A customer moves from regular repayment to early delinquency, such as 1–30 DPD, and shows signs of progressing towards more serious overdue categories. The system should identify the risk of continued deterioration based on repayment history, exposure and other available signals. It should distinguish customers who recover quickly from those showing repeated or worsening delinquency.

6. SME and Business Financial Stress

A business customer may experience declining sales receipts, irregular account activity, reduced operating balances or increasing debt obligations. The application should identify potential financial stress using available transaction, repayment and exposure data. Participants may introduce additional SME indicators where suitable synthetic data is available.

7. Missing or Incomplete Customer History

Some customers may have limited repayment records, missing transaction history or insufficient financial information. The application must distinguish missing evidence from an actual deterioration signal. It should not automatically classify a customer as high risk solely because information is unavailable.

8. Portfolio Concentration and Emerging Risk Clusters

Several customers within the same product, industry, region or portfolio segment may begin showing similar signs of financial deterioration. The system should identify concentrations of emerging risk and allow users to investigate whether the pattern is isolated or affects a broader customer segment.

Level 1: Identify, Analyse and Flag Early Credit Risk Signals
Suggested time: 0 - 3 hours
Objective: Build a working Early Warning System that ingests customer repayment and exposure information, identifies emerging risk indicators, calculates warning scores and presents customers requiring attention.

Required Capabilities
1. Customer and Credit Risk Data Ingestion
Build an application that allows users to upload or select customer and credit portfolio information. The application should support the following sample data:

- Customer profiles containing customer identifiers, customer type, employment or business category and portfolio classification.
- Loan and facility information containing product type, sanctioned amount, outstanding balance, tenure, instalment amount and repayment schedule.
- Repayment histories containing due dates, payment dates, amounts due, amounts paid and repayment status.
- Delinquency information containing Days Past Due, overdue balances and historical delinquency flags.
- Customer exposure information containing existing facilities, outstanding obligations, credit limits and utilisation.
- Financial or transaction indicators containing income, account inflows, outflows, average balances and cash-flow trends.
- Risk signal definitions containing thresholds, conditions, weights and warning classifications.

The application should accept CSV, Excel and JSON formats. It must validate uploaded records, identify missing or inconsistent information and establish relationships between customers, loan facilities, repayment histories and financial indicators.
2. Early Warning Signal Detection
Implement a mechanism that identifies emerging credit risk signals using configurable rules, statistical analysis, machine learning or a combination of approaches. At a minimum, the application must detect the following scenarios:

Scenario	Expected detection behaviour
Repeated repayment delays	Identify customers with recurring late or missed instalments
Temporary missed payments	Distinguish isolated delays from sustained repayment deterioration
Declining cash flow	Identify customers showing sustained reductions in income or account inflows
Increasing exposure	Detect rising financial obligations or credit utilisation
Missing customer history	Recognise insufficient information without inventing risk indicators

The system should consider changes over time rather than evaluating every data point independently. For example, a single missed instalment followed by immediate recovery should be assessed differently from repeated late payments over several months.
3. Customer Warning Score Calculation
Develop a mechanism that calculates an Early Warning Score for each customer. The score may use a numerical scale, such as 0–100, or categories such as Low, Medium, High and Critical. The scoring mechanism should consider relevant indicators, including repayment consistency, delinquency history, financial exposure, income or cash-flow deterioration and the number of active warning signals.
Participants must clearly define whether a higher score represents greater risk. The application must explain how the score was calculated and which factors contributed to the result.
4. Customer Risk Watchlist

Develop a watchlist displaying customers who require monitoring or intervention. The watchlist should show customer identifier, product or facility, outstanding exposure, warning score, risk category, contributing indicators and monitoring status. Users should be able to filter the watchlist by product, customer segment, risk level, delinquency status and portfolio. Selecting a customer should display the underlying repayment history, exposure information and warning indicators.

5. Basic Warning Review and Classification

Allow a risk analyst to inspect an early warning alert and record an initial decision. Supported decisions should include:

- Continue Monitoring
- Requires Investigation
- False Positive
- Escalate for Review

The analyst should be able to provide a reason for the decision. The application must preserve the original warning score and evidence even when the analyst changes the classification.

Expected Submission

A working application demonstrating data ingestion, early warning signal detection, customer risk scoring, an explainable watchlist and basic analyst review.

5. Customer Service Management (CSM) Voice Agent

Category: Conversational AI, Voice Automation & Customer Service Intelligence
Applicable Domains: Digital Lending, Loan Management, Core Banking, Insurance and Financial Services
Focus: Voice-Based Customer Assistance, Service Request Management, Customer Authentication, Workflow Automation, Multilingual Support and Human Escalation

Problem Statement

Financial institutions handle thousands of customer interactions involving account enquiries, loan repayments, application statuses, insurance policies, service requests, complaints and general product information. Despite significant investments in digital banking applications and customer service platforms, many customers continue to depend on call centres for routine enquiries and service requests.

A customer may call to understand the status of a loan application, enquire about an upcoming instalment, request a repayment statement, report a failed transaction or raise a complaint. These interactions often require customer service representatives to verify the customer's identity, navigate multiple financial systems, retrieve information, explain the outcome and create or update service requests.

Traditional Interactive Voice Response (IVR) systems typically rely on predefined menus and fixed interaction flows. Customers may need to navigate several options before reaching the appropriate service.

Although conversational AI can improve these experiences, simply building a voice-enabled chatbot is insufficient for financial institutions. The solution must understand customer intent, authenticate users, retrieve accurate information from connected financial systems, execute permitted service operations and maintain a complete record of the interaction.

For example, a customer may call and say:

"I applied for a personal loan last week. Can you tell me the status? Also, when will my first instalment be due?"

The system should recognise that the customer has two related questions, verify their identity, retrieve the relevant application and repayment information, and respond conversationally.

If the customer subsequently asks:

"Can you send me the repayment schedule?"

The agent should understand the context of the conversation and initiate the appropriate service request or document delivery workflow through an approved API. Similarly, if a customer reports an unauthorised transaction or expresses dissatisfaction with a previous service request, the agent should recognise the sensitivity of the situation and escalate appropriately. The challenge is to build an AI-powered Customer Service Management Voice Agent that can conduct natural spoken conversations with customers, understand their service requirements, securely access financial system information, execute authorised service workflows and escalate unresolved or sensitive interactions to human agents.

The solution must go beyond answering frequently asked questions. It should operate as a functional customer service agent integrated with simulated lending, banking, insurance and customer service systems. It must support multiple customer intents within the same conversation, retain conversational context, handle interruptions and clarification requests, and avoid disclosing sensitive information before appropriate customer verification. The application should provide both a voice interaction interface and an operational dashboard where customer service teams can inspect conversations, service requests, outcomes and escalations.

Participants are encouraged to introduce additional customer service automation capabilities, advanced voice experiences and innovative approaches beyond the

mandatory scenarios. The organiser will assess actual application behaviour rather than a prerecorded voice demonstration, static interface or standalone AI-generated response.

Customer Service Scenarios to Be Addressed

The solution should consider the following customer service scenarios. These represent practical interactions across lending, banking, insurance and other financial services.

1. Loan Application Status Enquiry

A customer contacts the financial institution to enquire about a submitted loan application. The voice agent should authenticate the customer, retrieve the application status from the simulated Loan Origination System (LOS) and explain the current stage in simple language.

For example, if an application is awaiting document verification, the agent should communicate this accurately rather than claiming that the loan has been approved.

2. Loan Repayment and Instalment Enquiry

A customer asks about the next instalment amount, due date, outstanding loan balance or repayment schedule. The agent should retrieve information from the simulated Loan Management System (LMS) and provide an accurate response. It should distinguish scheduled instalments from overdue amounts and avoid inventing financial values.

3. Account and Transaction Enquiry

A customer asks about an account balance, recent transaction or transaction status. The agent should retrieve the information from a simulated core banking service after successful authentication. If a transaction is pending or failed, the agent should explain the recorded status without making unsupported assumptions about the cause.

4. Service Request Creation and Tracking

A customer requests a loan statement, repayment schedule, account-related document or another supported service. The voice agent should identify the request, collect the required information, create a service ticket through the simulated CSM platform and provide a reference number. The customer should also be able to enquire about the status of an existing service request.

5. Complaint Registration and Escalation

A customer reports an unresolved issue, failed transaction, incorrect charge or unsatisfactory service experience. The agent should capture the complaint, classify it appropriately, create a complaint record and communicate the reference number. Where the issue is sensitive, urgent or outside the agent's authority, it should escalate to a human representative.

6. Insurance Policy and Claim Enquiry

A customer asks about an insurance policy, premium due date, coverage status or submitted claim. The agent should retrieve the relevant information from a simulated insurance system and respond based on the available records. It must not promise claim approval or coverage where the underlying policy information does not support that conclusion.

7. Multiple Requests Within One Conversation

A customer may ask about a loan repayment, then request a statement and subsequently enquire about an existing complaint. The agent should recognise each intent, maintain the correct customer context and handle the requests without unnecessarily restarting the conversation.

8. Interrupted, Ambiguous and Incomplete Conversations

Customers may interrupt the agent, change their question, provide incomplete information or use informal language.

For example:

"No, I meant my vehicle loan, not my personal loan."

The agent should recognise the correction, update the conversation context and continue appropriately.

9. Sensitive Requests and Authentication Failures

A customer may fail identity verification, request another person's financial information or ask the agent to perform an unauthorised action. The system should enforce access restrictions, avoid disclosing protected information and escalate when required.

10. Multilingual Customer Assistance

Customers may prefer communicating in English, Hindi, Arabic or another supported language. The application should support natural spoken interactions in at least one language at Level 1, with multilingual capability introduced progressively.

Participants are encouraged to demonstrate additional languages relevant to the financial institution's operating markets.

Level 1: Understand Customer Voice Requests and Provide Accurate Assistance
Suggested time: 0 - 3 hours
Objective: Build a working voice-enabled customer service application that understands spoken customer enquiries, identifies intent, retrieves information from simulated financial systems and provides accurate spoken responses.

Required Capabilities
1. Voice Input and Speech Recognition
Build an application that allows customers to interact using spoken language.
The application should capture customer speech through a microphone or equivalent voice interface and convert it into text for processing. It should support natural customer questions rather than requiring fixed commands. For example, the system should recognise variations such as:

- "When is my next EMI?"
- "How much do I need to pay next month?"
- "Tell me my next loan instalment."
- "What's the due date for my personal loan?"

The application should identify that these questions relate to repayment information. The interface must display the recognised transcript so that users can inspect whether the customer's speech was understood correctly.
2. Customer Intent Identification
Develop a mechanism that classifies customer requests into appropriate service categories. At a minimum, the application should recognise:

Customer intent	Expected behaviour
Loan application enquiry	Retrieve and explain application status
Repayment enquiry	Retrieve instalment amount and due date
Account enquiry	Retrieve permitted account information
Service request enquiry	Retrieve existing ticket status
General product enquiry	Provide information from an approved knowledge source
Complaint or issue	Identify the issue and initiate the appropriate service flow

The system should recognise different ways of expressing the same request. If the customer's intention is unclear, the agent should ask a clarification question rather than selecting an unrelated service.
3. Customer and Financial System Data Integration
Connect the application to simulated financial system APIs or organiser-provided datasets. The application should support:

- Customer profiles containing identifiers and basic customer information.
- Loan applications containing application references, product details and current statuses.
- Loan accounts containing repayment schedules, outstanding balances and instalment details.
- Bank accounts containing permitted balance and transaction information.
- Service requests containing ticket identifiers, categories and statuses.
- Product information and FAQs containing approved service explanations.

Participants may use mock APIs or locally hosted services to simulate the underlying systems. The application must retrieve actual values from the connected data rather than generating fictional account balances, loan amounts or application statuses.
4. Basic Customer Verification
Implement a basic customer verification mechanism before disclosing account-specific information. General product questions may be answered without authentication.

However, sensitive enquiries involving loan balances, transactions, account details or personal service records must require successful verification.
The system should prevent unauthenticated users from accessing protected customer information.

5. Natural Spoken Responses

Convert the agent's responses into speech so customers can hear the information. The responses should be conversational, concise and understandable. For example:
"Your personal loan application is currently under document verification. No additional action is recorded as required from you at this stage." The voice agent should avoid reading lengthy technical responses or raw API data. The application should also display the text response alongside the spoken output.

6. Basic Customer Service Interface

Develop an interface showing the current voice conversation. The interface should display:

- Recognised customer speech.
- Identified customer intent.
- Authentication status.
- Retrieved service information.
- Agent response.
- Conversation history.

Users should be able to start and end a conversation and inspect the information used to generate a response.

Expected Submission

A working voice-enabled customer service application demonstrating speech recognition, intent identification, customer verification, financial information retrieval and spoken responses.

6. Collections Voice Agent

Category: Conversational AI for Collections, Loan Servicing & Responsible Customer Engagement
Applicable Domains: Digital Lending, Loan Management, Retail Finance, SME Finance and Core Banking
Focus: Repayment Assistance, Customer Engagement, Promise-to-Pay, Dispute Handling, Collections Workflows and Responsible Voice Automation

Problem Statement

Financial institutions spend considerable time and effort contacting customers about upcoming, overdue and missed loan repayments. Before each conversation, collections agents must review repayment histories, outstanding amounts, previous customer commitments, disputes and current account status.

Traditional outbound collections often depend on manual calling, rigid scripts and inconsistent follow-up. Customers may miss payments for very different reasons, including temporary cash-flow difficulties, salary delays, payment processing errors or genuine financial distress. Treating every customer identically can reduce collections effectiveness, damage customer relationships and create conduct risk for the institution.

For example, one customer may have already paid through a branch but the payment has not yet been reflected in the loan account. Another may be willing to pay next week after receiving a delayed salary. A third may have recently lost their job and require specialist hardship assistance. A scripted reminder that repeats the overdue amount to all three customers serves none of them well.

The challenge is to build an AI-powered Collections Voice Agent that conducts responsible, personalised repayment conversations, retrieves accurate loan information, understands customer responses and records agreed follow-up actions. The agent should support both inbound calls and simulated outbound calls, distinguish payment reminders from overdue collections, capture promises to pay, recognise disputes and hardship, and escalate situations requiring human intervention.

The application must not merely speak a scripted reminder. It should interact with a simulated Loan Management System, create collection activity records, validate customer commitments against account information and maintain a complete history of conversations and decisions.

The solution must protect customer dignity and privacy at all times. It must authenticate the customer before discussing account-specific details, avoid coercive or misleading language, and ensure that consequential actions remain under human control.

Participants are encouraged to introduce additional collections intelligence capabilities and innovative approaches beyond the specified scenarios. Solutions demonstrating meaningful improvements in repayment engagement while preserving responsible conduct will receive additional consideration within the evaluation framework.

The organiser will assess actual application behaviour rather than a prerecorded voice demonstration, static interface or standalone AI-generated response.

Collections Scenarios to Be Addressed

The solution should consider the following collections scenarios. These represent practical repayment conversations faced by lenders and provide the basis for progressive evaluation.

1. Upcoming Repayment Reminder

A customer has an instalment due within the next few days. The agent should inform the customer about the approaching due date and amount, confirm that the details are understood and answer simple questions about payment channels. The conversation should be informative rather than framed as an overdue collection.

2. Missed Instalment Follow-Up

A customer has missed a scheduled instalment. The agent should explain the overdue amount and the original due date, ask whether the customer intends to make payment and capture the outcome. It must communicate only values recorded in the loan system and must not invent penalties or charges.

3. Promise-to-Pay and Partial Payment

A customer agrees to pay on a future date or can pay only part of the overdue amount. The agent should capture the proposed amount and date, validate them against the account information and permitted rules, and create a follow-up record. It should distinguish an agreed commitment from a proposal requiring supervisor approval.

4. Payment Already Made and Disputes

A customer states that the payment has already been made or disputes the overdue amount. The agent should check the available transaction records and, if the payment cannot be confirmed, create a dispute case rather than repeatedly requesting payment or implying that the customer is being untruthful.

5. Financial Hardship and Vulnerable Customers

A customer reports job loss, illness, bereavement or other circumstances affecting their ability to pay. The agent should recognise the situation, avoid pressuring the customer into an unaffordable commitment and escalate the case to a trained specialist.

6. Broken Promise-to-Pay

A customer previously committed to a payment that was not fulfilled. The agent should recognise the earlier commitment from the collections history, acknowledge it appropriately and recommend a suitable next action based on configured collections rules.

7. Wrong Party Contact and Failed Verification

The person answering the call may not be the borrower, or the customer may fail verification. The agent must not disclose the existence of a loan, outstanding amounts or other sensitive information to unauthorised individuals.

Level 1: Understand Repayment Status and Conduct Voice Conversations
Suggested time: 0 - 3 hours
Objective: Build a working voice application that retrieves repayment information, authenticates customers and conducts accurate, natural repayment conversations.

Required Capabilities
1. Loan and Collections Data Integration
Build an application that allows users to upload or select synthetic collections data or connect to mock loan management APIs. The application should support the following sample data:

- Customer profiles containing customer identifiers, contact details, preferred language and verification attributes.
- Loan accounts containing product type, outstanding principal, instalment amount, repayment schedule and account status.
- Overdue records containing missed instalments, overdue amounts, Days Past Due and applicable charges.
- Previous collections activities containing earlier calls, outcomes, commitments and disputes.

The application should accept CSV, Excel and JSON formats. It must validate customer and loan identifiers, identify missing records and flag inconsistent repayment amounts rather than silently using incorrect values.
2. Voice Interaction and Intent Recognition
Provide real voice input and spoken responses. The application should convert customer speech into text, identify the customer's intent and respond in natural spoken language. At a minimum, the application must recognise the following intents:

Customer intent	Expected behaviour
Repayment enquiry	Retrieve and explain the instalment amount, due date or overdue balance
Payment commitment	Capture the proposed payment date and amount for confirmation
Payment already made	Check transaction records before responding
Inability to pay	Acknowledge the difficulty and identify the appropriate next step
Request for clarification	Repeat or simplify the relevant information
Request for human assistance	Initiate the escalation flow

The interface must display the recognised transcript so that reviewers can inspect whether the customer's speech was understood correctly. Where the intent is unclear, the agent should ask a clarification question rather than selecting an unrelated response.
3. Customer Verification
Implement a mock customer authentication step before disclosing account-specific information. General service information, such as available payment channels, may be provided without verification. Loan balances, overdue amounts and repayment history must only be discussed after successful verification. Failed verification attempts must be recorded.
4. Accurate Repayment Communication
Retrieve and communicate the correct instalment amount, due date, overdue amount and repayment status from the connected data. For example: "Your instalment of ₹9,800 was due on 5 October and is currently shown as unpaid." The agent must not invent balances, penalties, settlement offers or payment deadlines. The text response should be displayed alongside the spoken output.
5. Basic Interaction Dashboard
Develop a dashboard showing the current and completed conversations. The dashboard should display the transcript, recognised intents, verification status, retrieved repayment

information, interaction outcome and call status. Users should be able to open any interaction and inspect the information used to generate each response.

Expected Submission
A working voice agent that authenticates customers, retrieves actual repayment records, communicates accurately, recognises customer responses and stores interaction history.

7. Claims Leakage Detection Agent

Category: AI for Insurance Claims Intelligence & Financial Leakage Prevention
Applicable Domains: General Insurance, Health Insurance, Motor Insurance and Claims Management
Focus: Claims Validation, Duplicate Detection, Overpayment Prevention, Policy Compliance, Investigation and Settlement Review

Problem Statement

Insurance companies process large volumes of claims involving policy coverage, supporting documents, invoices, service providers, claim histories and settlement calculations. Each claim must be checked against the applicable policy terms before an amount is approved for payment.

Financial leakage occurs when claims are paid incorrectly. This may result from duplicate submissions, excessive charges, incorrect calculations, missed policy exclusions, inappropriate benefits or simple processing errors. Individually, these amounts may appear small, but across a large claims portfolio they can represent a significant and avoidable financial loss.

Not every leakage event is fraudulent. Many result from operational mistakes, incomplete information or inconsistent application of policy rules. A claims handler under time pressure may overlook a deductible, approve an invoice line that was already paid under an earlier claim or miss that a treatment falls under a policy exclusion.

For example, a health insurance claim may include a hospital invoice in which the same diagnostic test appears twice, a room charge that exceeds the policy's daily limit and a consumable item that the policy explicitly excludes. Each issue may be minor on its own, but together they could cause the insurer to overpay by a material amount.

The challenge is to build an AI-powered Claims Leakage Detection Agent that examines insurance claims, identifies potential financial leakage, explains the supporting evidence and enables claims teams to investigate and correct questionable outcomes before settlement. The solution should combine structured claims data, policy conditions, supporting documents and historical claim information.

The application must distinguish confirmed rule violations from suspicious patterns requiring human review. It should calculate the potential leakage amount wherever it can be determined, explain why a claim was flagged, retain the supporting evidence and allow authorised users to confirm, correct or dismiss each finding.

Participants are encouraged to introduce additional leakage detection techniques and innovative approaches beyond the specified scenarios. Solutions demonstrating meaningful detection of complex or emerging leakage patterns will receive additional consideration within the evaluation framework.

The organiser will assess actual application behaviour rather than a static dashboard, predefined presentation or standalone AI-generated response.

Claims Leakage Scenarios to Be Addressed

The solution should consider the following types of claims leakage. These scenarios represent practical challenges faced by insurers and provide the basis for progressive evaluation.

1. Duplicate Claims and Invoice Line Items

The same incident, invoice or charge may be submitted more than once, either within a single claim or across separate claims. The application should identify exact duplicates as well as near-duplicates with minor differences in dates, amounts or reference numbers.

2. Policy Limit Breaches and Excluded Services

A claim may request an amount that exceeds an applicable coverage limit or include expenses that are not covered under the policy. The system should evaluate each claim line against the relevant policy terms and identify the specific limit or exclusion involved.

3. Incorrect Settlement Calculation
Deductibles, co-payments, sub-limits and benefit caps may be applied incorrectly. The application should recalculate the expected settlement amount using the supplied policy rules and highlight any differences from the proposed settlement.
4. Unusual Provider Billing and Related-Party Claims
A provider's charges may be materially inconsistent with comparable claims, or several claims may share unusual combinations of claimant, provider, invoice or incident information. The system should identify these patterns for review without automatically concluding that they are fraudulent.
5. Legitimate Lookalikes and Incomplete Evidence
Similar claims may represent separate valid incidents, such as repeated physiotherapy sessions, and must not automatically be rejected. Equally, some claims cannot be assessed reliably because essential information is missing. The application should request clarification rather than inventing a finding.